

RQ1: Profile accuracy and abstention

Research question

RQ1. Can the pipeline construct accurate site profiles, and does it abstain where a site is silent rather than guessing?

The analysis compares the final evidence-backed profiles against hand-built ground truth at three sites. It evaluates populated values, correct and false abstentions, unsupported values, evidence conflicts, and citation integrity.

Run configuration

- **60 completed profile builds:** 20 each for Anvil, Stampede3, and ND CRC.
- **Models:** GPT-5 Mini, GPT-5.6 Terra, Gemini Flash, and Gemini Pro.
- **Retrieval:** one BM25 run, three LLM-expanded BM25 runs, and one full-corpus run per site-model pair.
- **Frozen inputs:** every run for a site uses the same login measurement, pilot result, and documentation corpus. No evaluation run performs new discovery, measurement, or pilot submission.
- **Ground truth:** 147 fields for Anvil, 159 for Stampede3, and 68 for ND CRC, producing 7,480 run-field observations.
- **Balancing:** BM25 and full corpus have weight one; each expanded repetition has weight one third. Every model-retrieval configuration therefore has equal aggregate weight.
- **Matching:** populated values use field-specific numeric, set, pattern, or normalized-text rules. Site silence and not-applicable states are both evaluated as expected absence in this preliminary analysis.

The analysis is reproducible with:

```
python evaluation/rq1/run_rq1_analysis.py
```

Field outcomes

Site	Precision	Recall	Abstention accuracy	Field accuracy
Anvil	0.977	0.867	0.969	0.889
Stampede3	0.957	0.906	0.935	0.912
ND CRC	0.990	0.764	0.972	0.777
Macro mean	0.975	0.846	0.959	0.859

Precision exceeds recall in the macro result and at all three sites. The predicted abstention rate is highest at Anvil (0.306), followed by ND CRC (0.274) and Stampede3 (0.238).

Configuration fields are consistently strongest. Policy accuracy is lower: 0.821 at Anvil, 0.876 at Stampede3, and 0.809 at ND CRC. This is the portion of the profile most dependent on documentation extraction.

Evidence behavior

The profiles expose configuration-versus-policy disagreement. Stampede3 produces six walltime conflicts per balanced configuration. Anvil produces 7.33 of its eight expected walltime conflicts on average; one configuration failed to recover the documented limits. Anvil also contains 0.28 other conflicts per configuration, primarily an erroneous GPU-count disagreement in a few runs.

GPU facts demonstrate both recovery and difficulty. Stampede3 recovers the H100 and PVC counts and models in every configuration. Anvil recovers the A100 model for the `ai` and `gpu` partitions in 0.222 of configurations and for `gpu-debug` in 0.028.

Citations

All 3,052 accepted citations are mechanically valid: the cited chunk exists, belongs to the target site, uses the frozen source URL, and contains the quoted text. The 2,257 accepted findings all contain at least one citation. This does not establish semantic entailment; that requires a separate manual audit.

Answer to RQ1

The pipeline constructs mostly accurate and conservative profiles for the three evaluated sites. Macro precision is 0.975, recall is 0.846, abstention accuracy is 0.959, and field accuracy is 0.859. Precision exceeds recall at every site, so the dominant failure is leaving a known value unresolved rather than populating an unsupported value.

The evidence paths are complementary. Login measurement directly establishes 30--38% of evaluated fields per site, documentation recovers another 22--39%, and pilot evidence contributes 4--10%. Correct abstention accounts for 6--21%, leaving 9% unresolved or wrong at Stampede3, 11% at Anvil, and 22% at ND CRC.

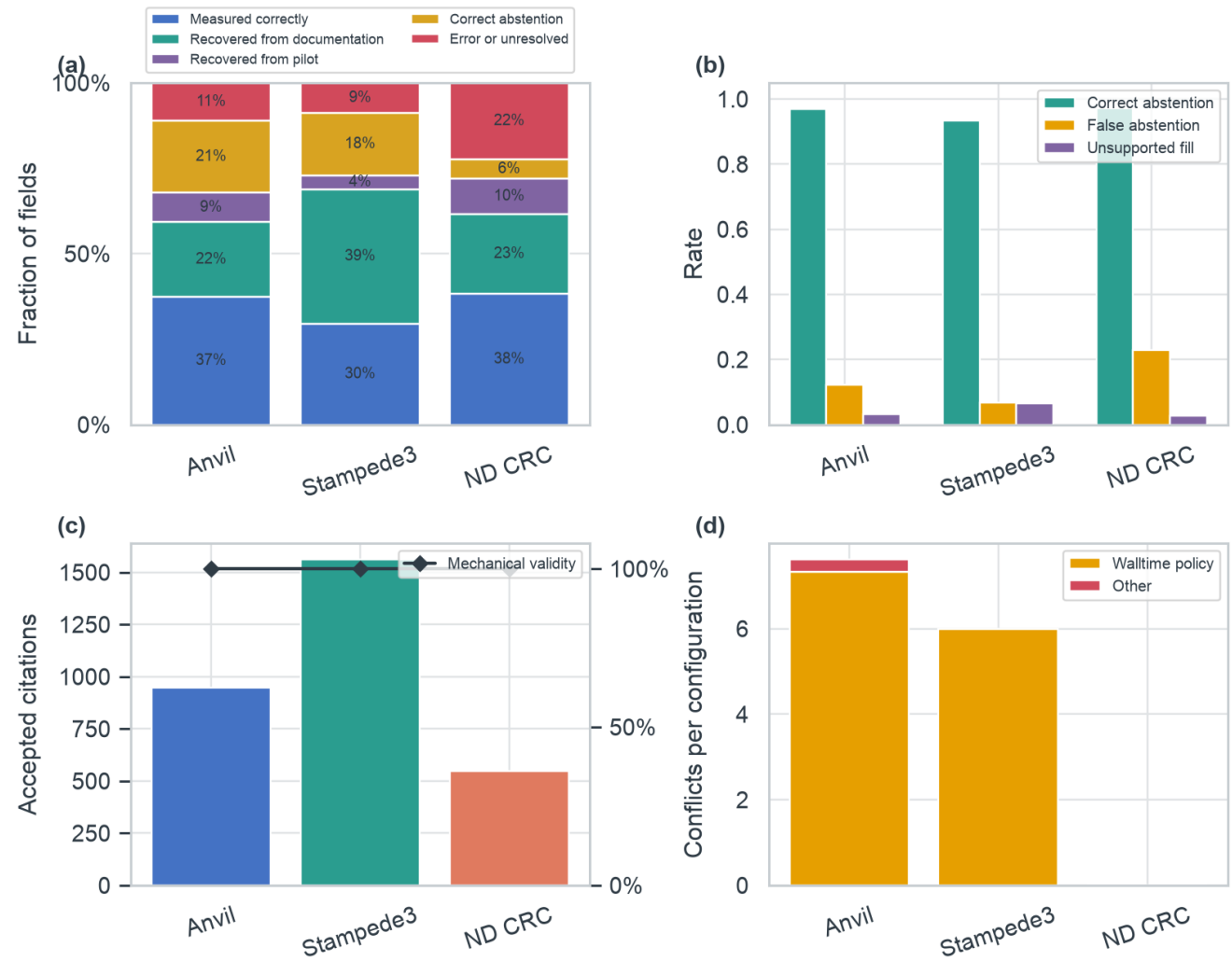
The result is site-dependent. Stampede3 reaches 0.912 field accuracy, Anvil 0.889, and ND CRC 0.777. ND CRC's lower recall shows that abstention is not always correct: some documented values remain unresolved. At the same time, the pipeline identifies measurement-documentation disagreements, including the Slurm walltime conflicts, and recovers GPU facts that login-node measurement misses. RQ1 is therefore answered positively for accuracy and conservative behavior, but not as complete coverage: profiles remain partial where discovery, extraction, or evidence is insufficient.

Figures

All figures use balanced configuration weights: BM25 and full corpus have weight one, while each of the three expanded-retrieval repetitions has weight one third.

The palette uses medium-saturation blue and teal for correct outcomes, red and amber for errors, and a colorblind-distinguishable blue, teal, and coral site encoding. It is designed for clear reproduction in an academic paper.

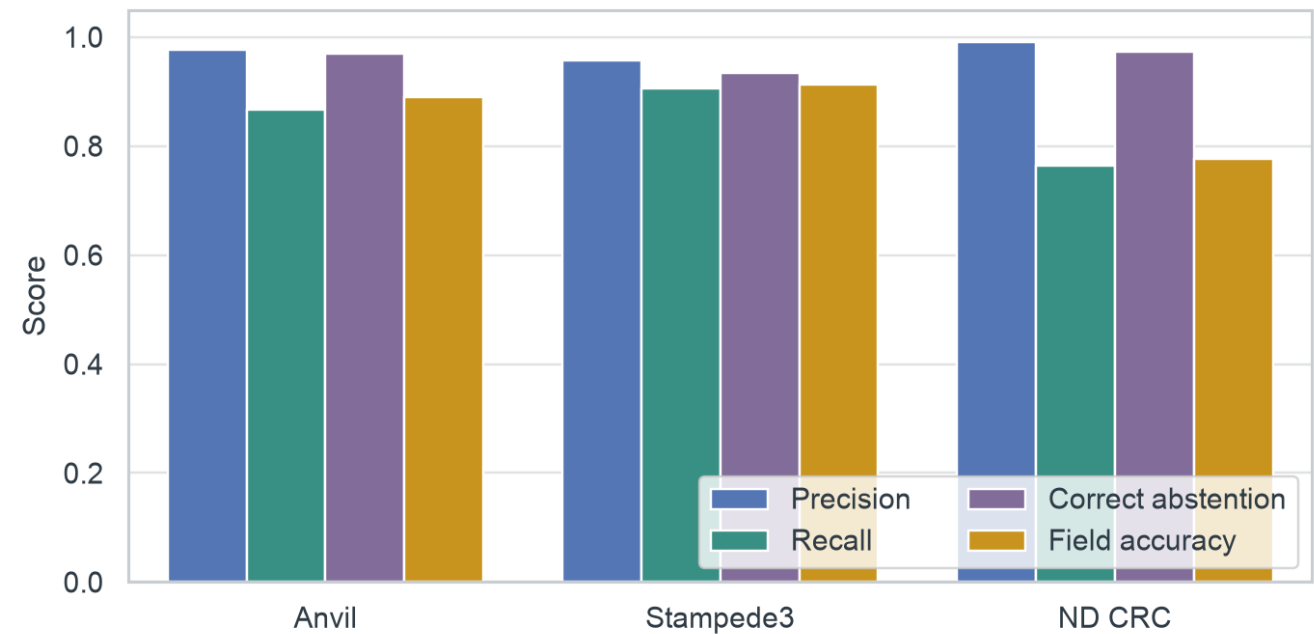
Compact RQ1 summary



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Panel (a) shows how login measurement, documentation, pilot evidence, and correct abstention contribute to each completed profile; the remaining segment is wrong or unresolved. Panel (b) separates correct abstention, false abstention, and unsupported fills. Panel (c) shows accepted citation volume and mechanical validity. Panel (d) shows explicit evidence conflicts, separating walltime-policy disagreements from other conflicts. Precision, recall, and field accuracy are omitted because they are reported in the table above.

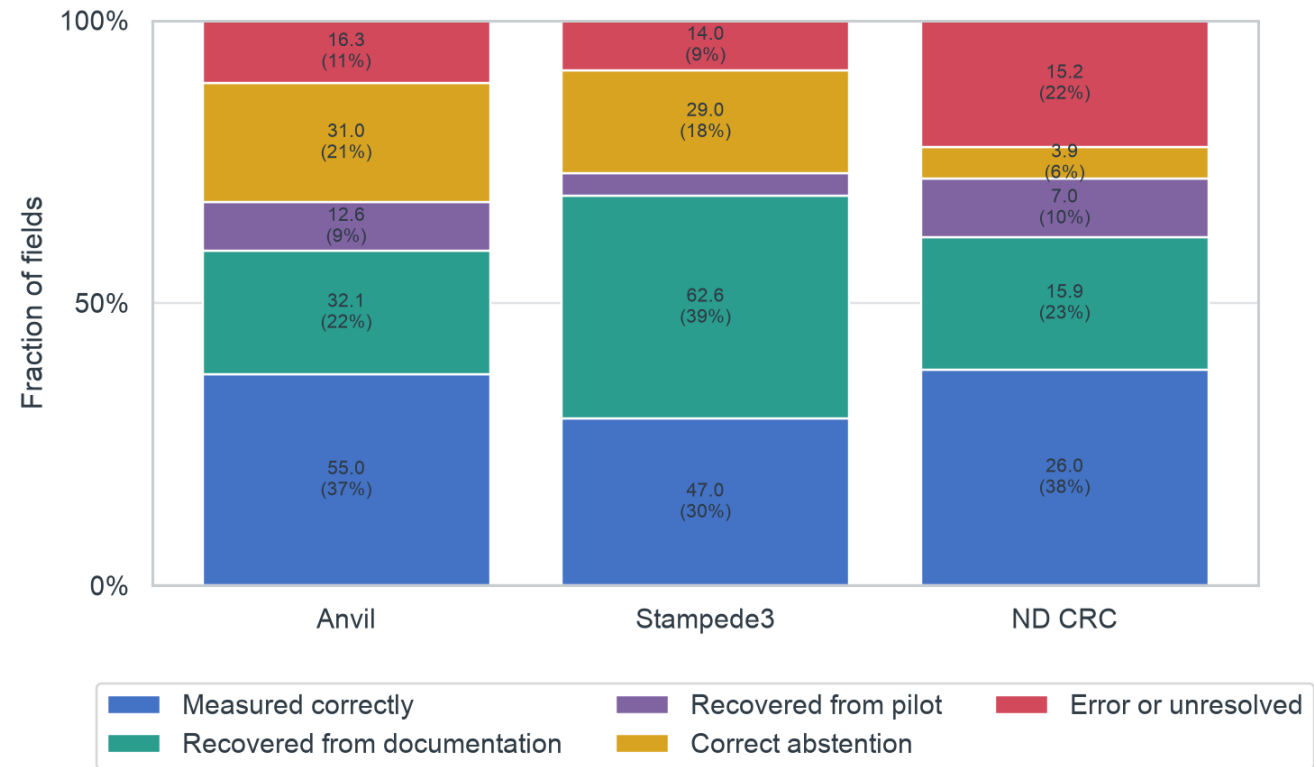
Site metrics



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This grouped comparison separates precision, recall, abstention accuracy, and overall field accuracy by site. ND CRC has the highest precision but the lowest recall because it leaves more ground-truth values unresolved.

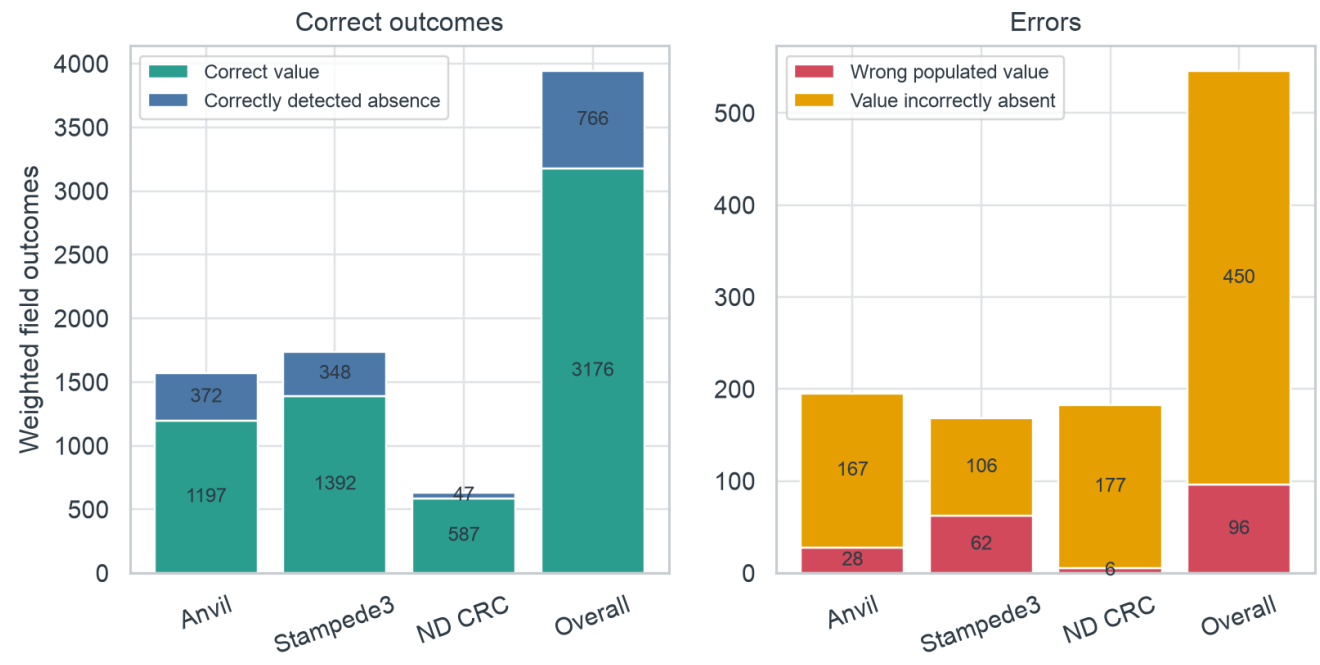
Profile completion waterfall



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Each bar accounts for every evaluated field through the first evidence path that establishes its correct final value: login measurement, documentation, or pilot evidence. Correct absence is separate, and the final segment combines wrong values with unresolved known values. Labels report the mean number of fields per balanced profile and the corresponding fraction of evaluated fields.

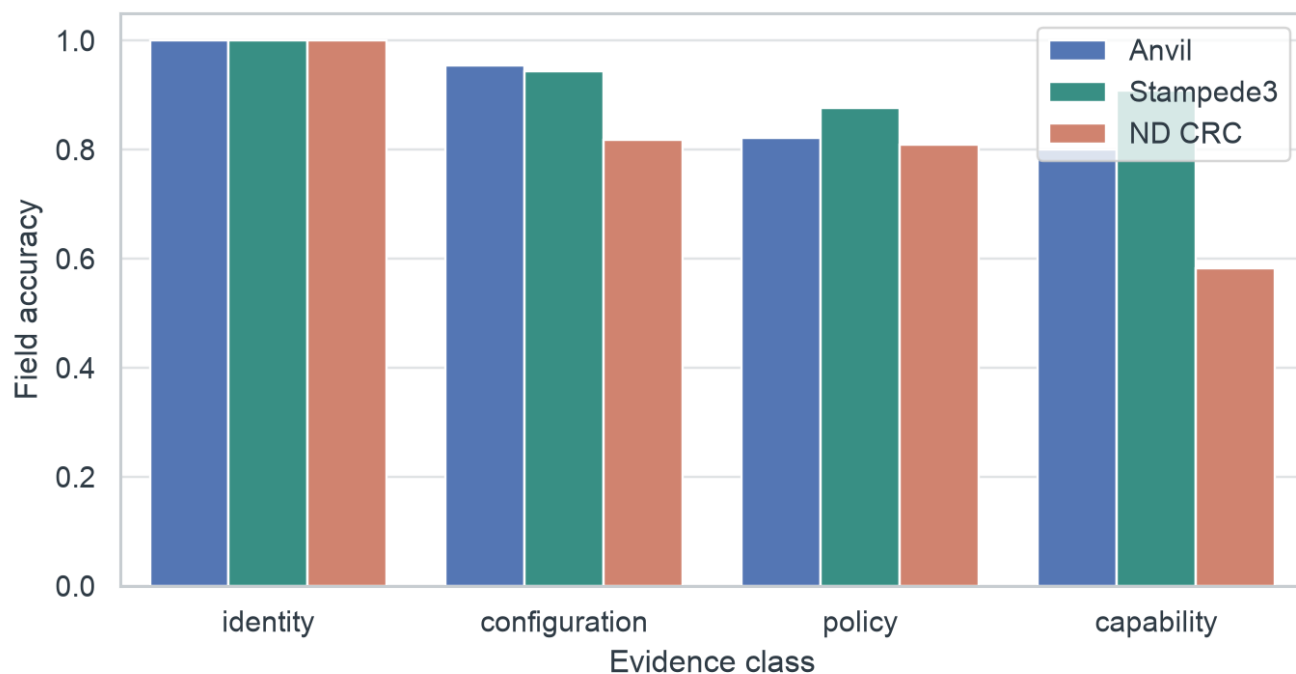
Combined outcome counts



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The left panel gives weighted counts of correct values and correctly detected absences. The right panel gives wrong populated values and values that should have been populated but were absent.

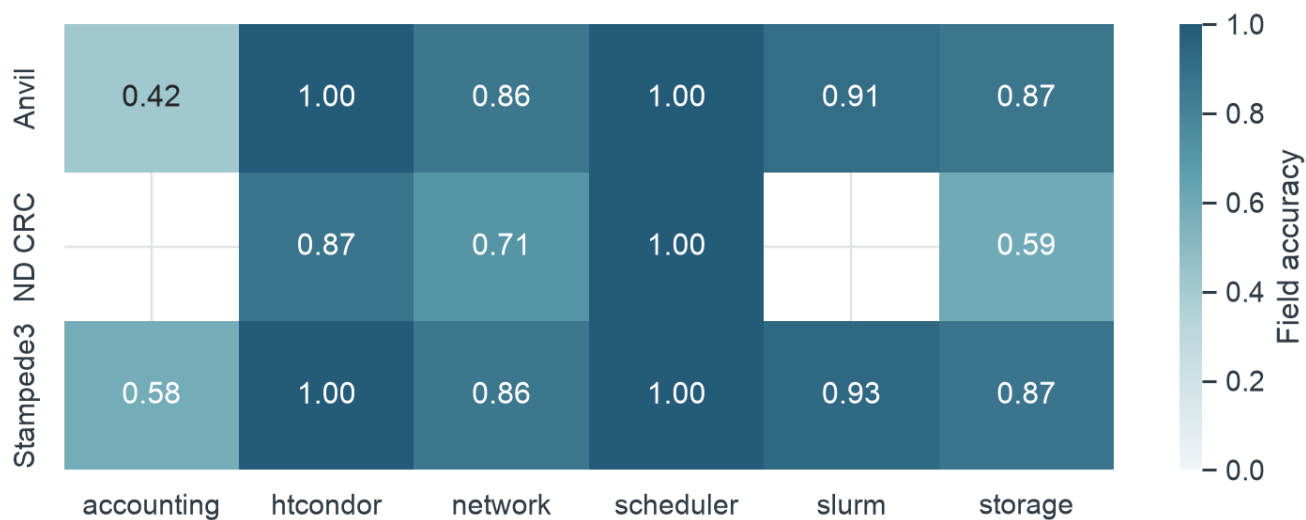
Accuracy by evidence class



[PDF](#)

This figure separates identity, configuration, policy, and capability fields. It shows where the full pipeline is strongest and where documentation extraction or pilot coverage still limits accuracy.

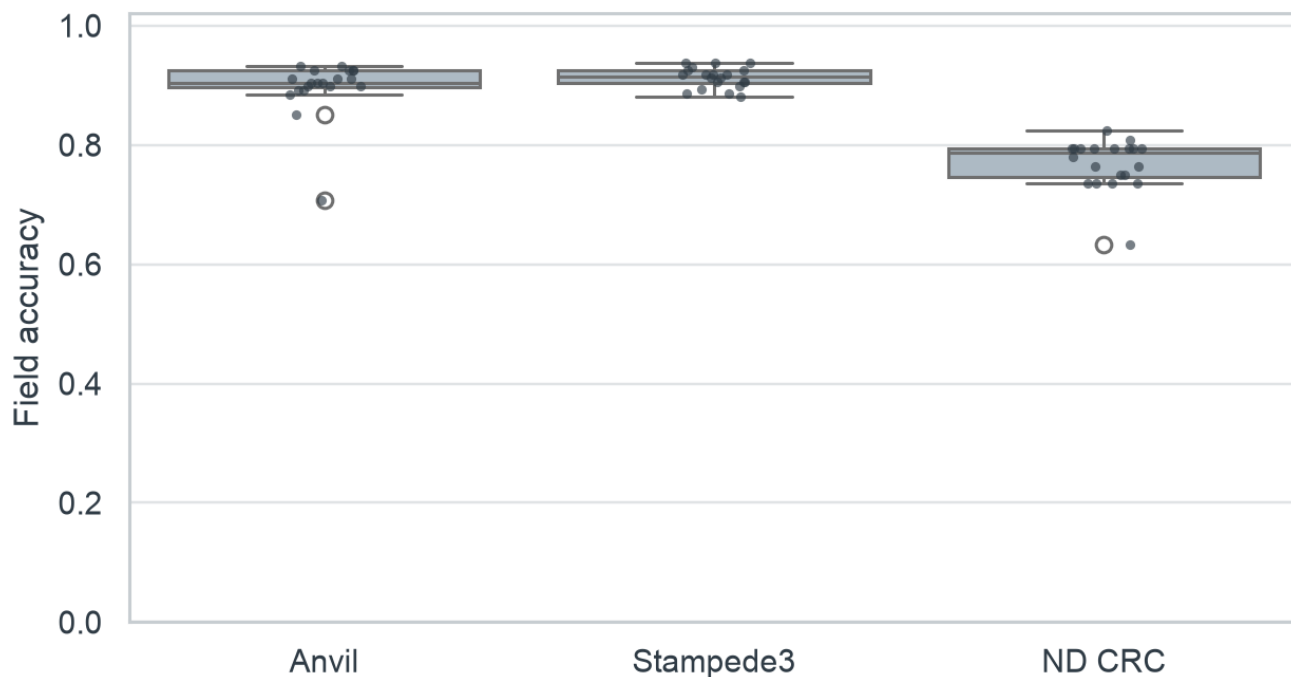
Accuracy by profile section



[PDF](#)

The heatmap localizes accuracy by site and profile section, making weaknesses in scheduler, storage, networking, or accounting fields visible without averaging them into one site-level score.

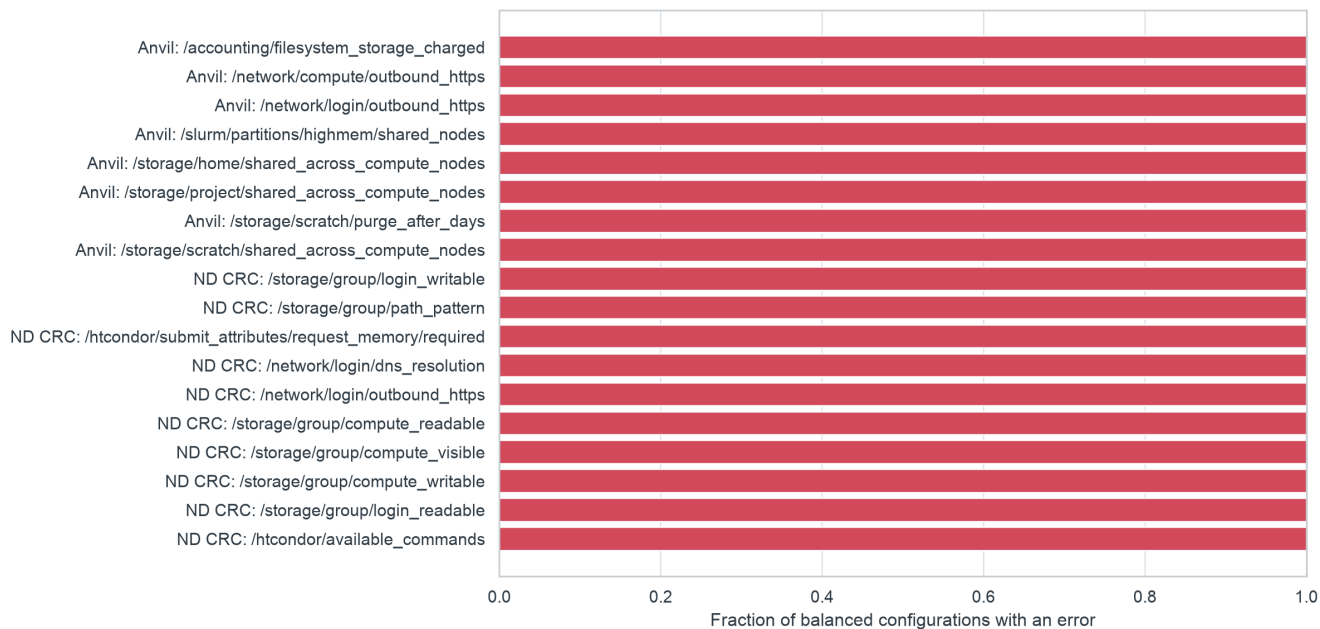
Run accuracy distribution



[PDF](#)

Each point is one model-and-retrieval run; the box summarizes the distribution for a site. The spread measures output variance that is hidden by aggregate precision and recall.

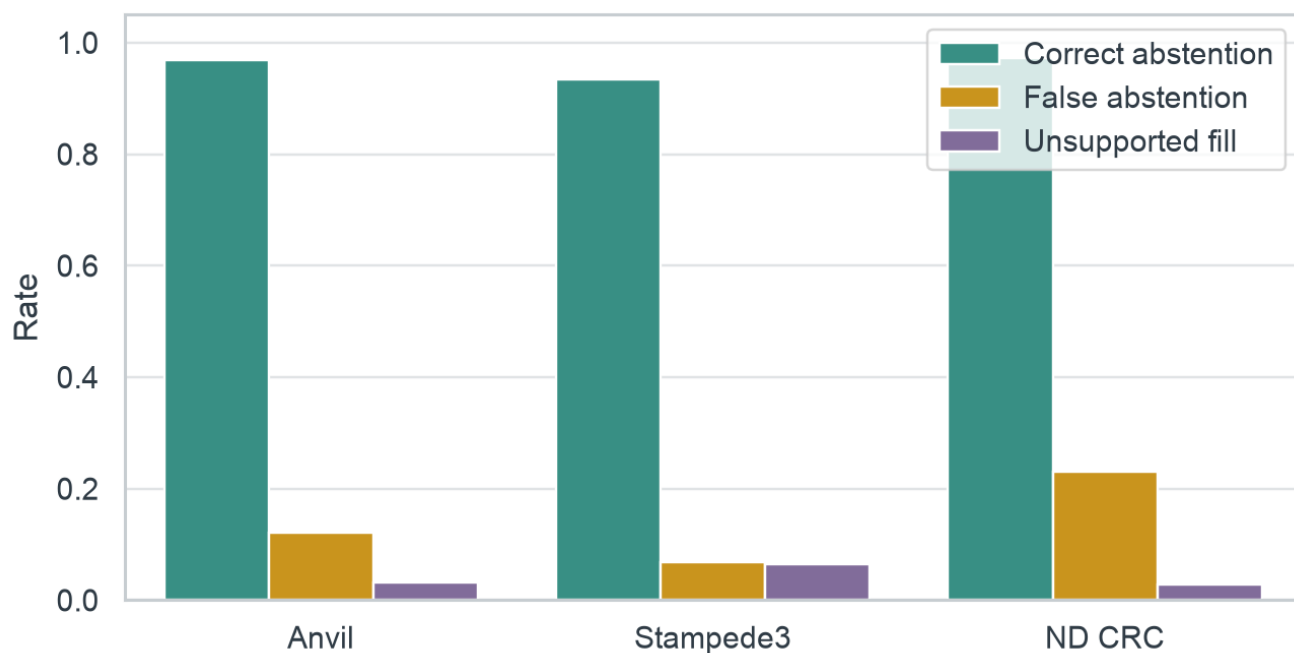
Most frequent field errors



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These are the 18 fields with the highest balanced error rate. An error includes a wrong value, a false abstention, or a value populated where ground truth says the field is absent.

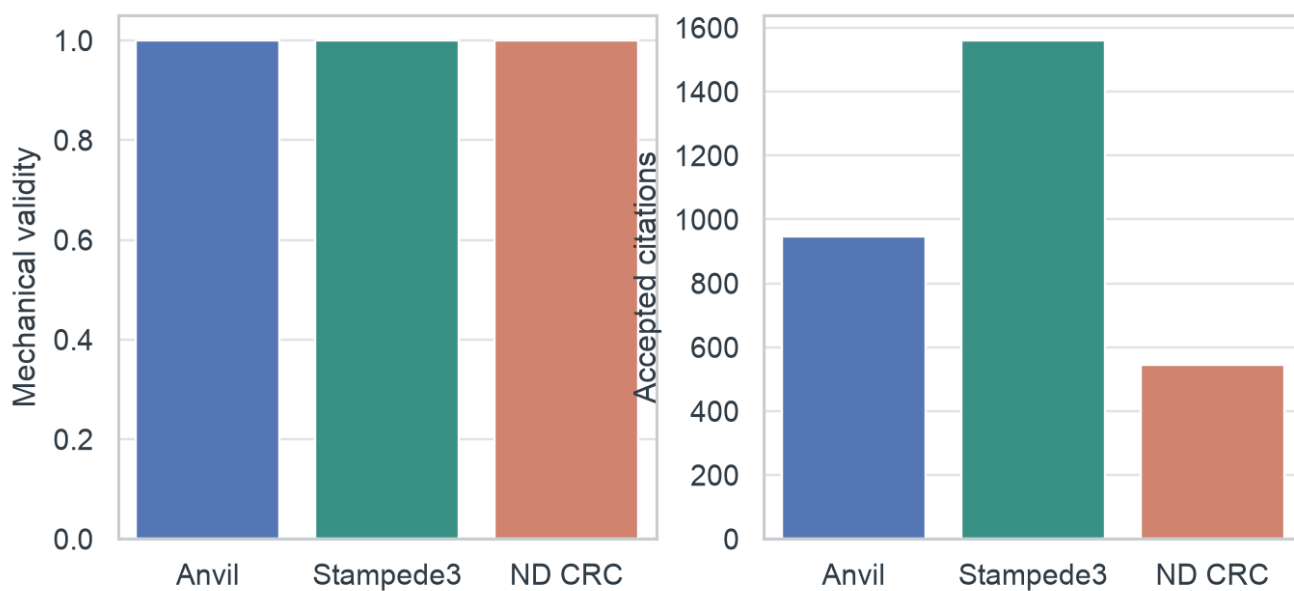
Abstention behavior



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Correct abstention measures silence on truly absent fields. False abstention measures missed known values, while unsupported fill measures values proposed for absent fields. These rates use different denominators and should not be read as parts of one stacked total.

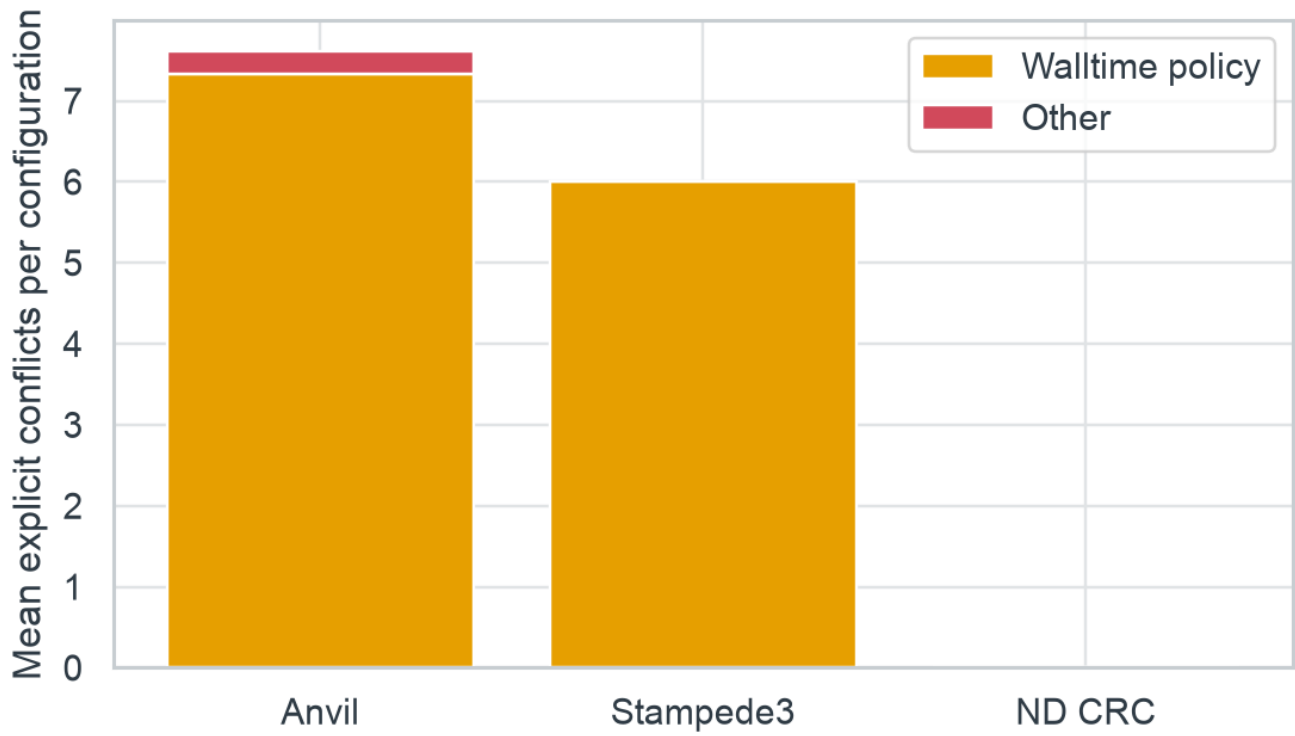
Citation integrity



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The left panel checks that citations point to stored target-site chunks and contain the recorded quote. The right panel shows citation volume. Mechanical validity does not by itself prove that the quote semantically entails the field.

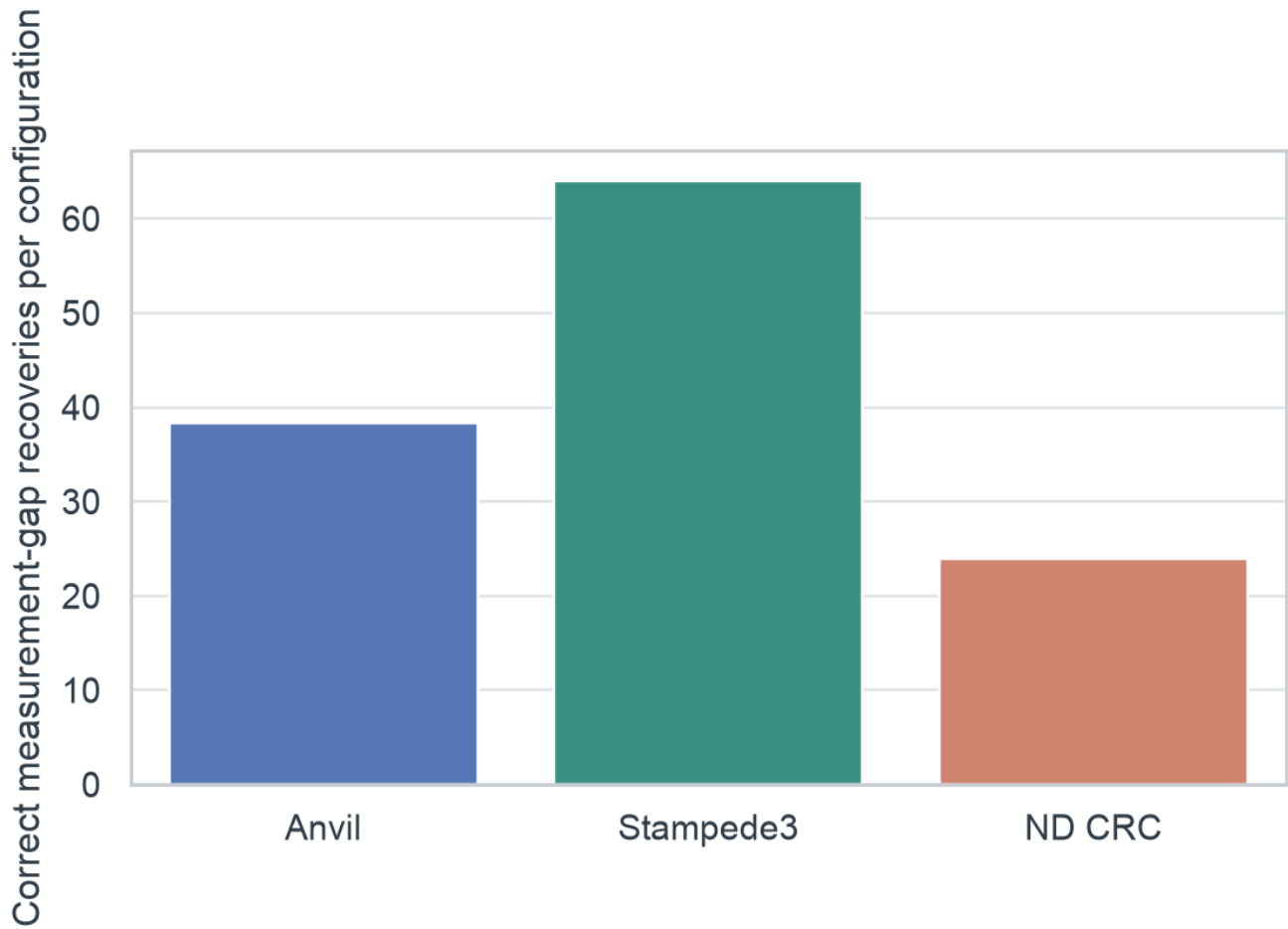
Source conflicts



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This figure counts explicit evidence disagreements per balanced configuration. Walltime conflicts are separated because they capture the central case where visible scheduler configuration reports unlimited time but documentation states an enforced policy limit.

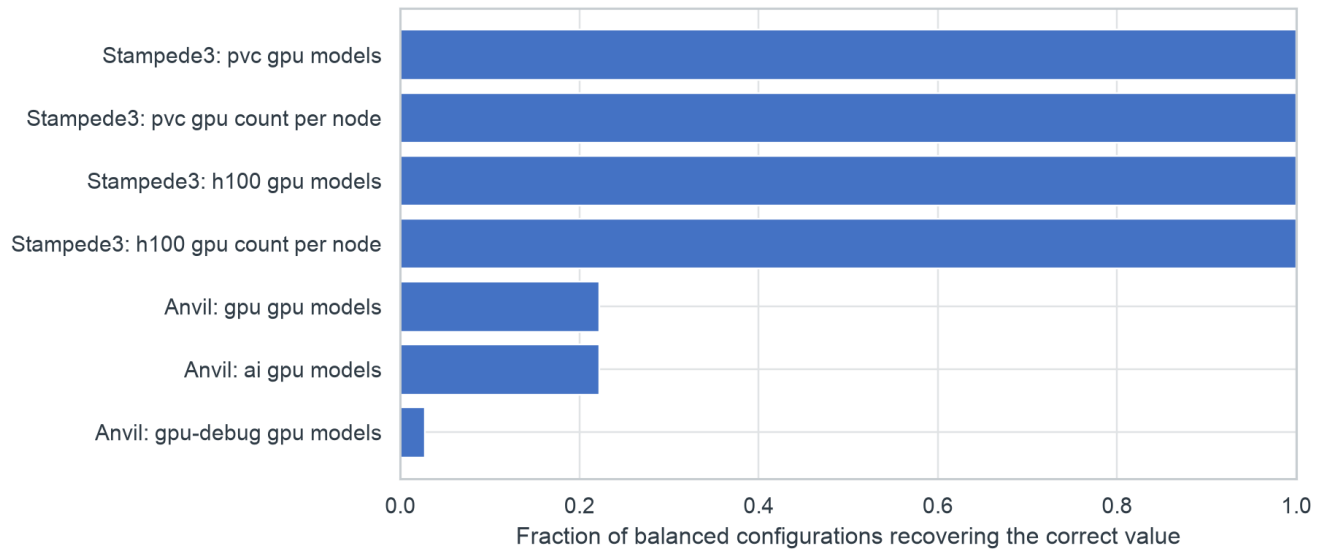
Measurement-gap recovery



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The bars count correct values recovered per configuration when login measurement supplied no value. They quantify what documentation extraction and pilot evidence add beyond measurement alone.

GPU measurement-gap recovery



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This field-level view reports how often the pipeline recovers the correct GPU count or model when login-node measurement cannot see it. Stampede3 is consistently recovered, while the Anvil GPU-model fields remain difficult.

Interpretation boundary

These numbers establish what the current evaluator measures. Claims about semantic citation validity and why a model made an error require manual review. Model and retrieval comparisons belong to RQ2 and are not inferred from the site-level RQ1 aggregates.